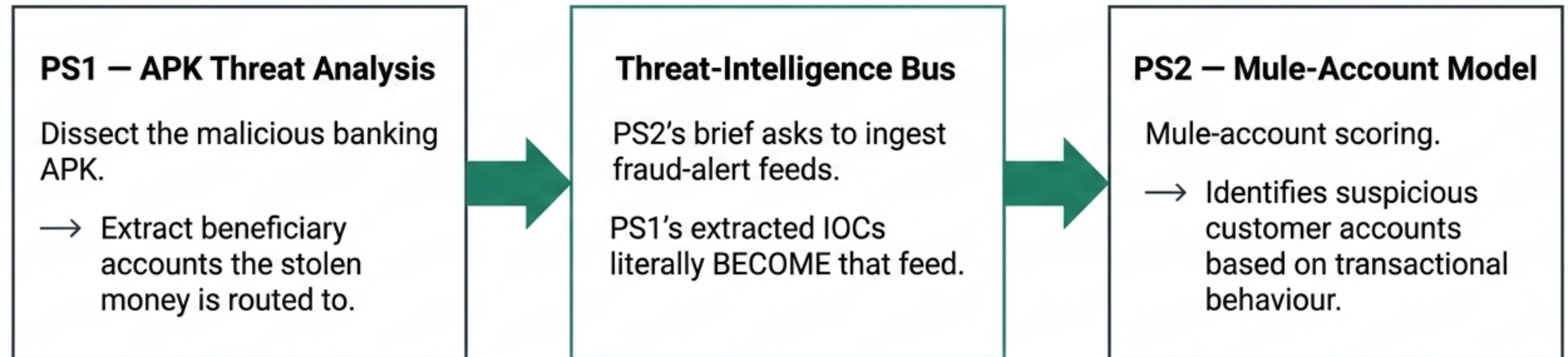


Two problem statements joined into ONE fraud kill-chain.

CyberShield | Team: Burning Platform | Problem Statements 1 & 2, combined



Each service stands alone and fully answers its own statement; the fusion is the upside.

PS1: GenAI APK Threat Analysis & Risk Scoring

**GenAI interprets and writes the report — it never decides the verdict;
resistant to prompt injection by design.**

Deterministic Engine (The Verdict)

The core scoring mechanism that banks can audit.

- ✓ Evidence-weighted risk score based strictly on static/dynamic code features.
- ✓ 4 auditable YARA-X behavioural rules operating directly over DEX bytecode.
- ✓ Deterministic IOC extraction (C2 URLs, Telegram bot tokens, beneficiary UPI handles).

GenAI Layer (The Interpretation)

The contextual wrapper that saves analyst time without compromising safety.

- ✓ Reads extracted evidence only.
- ✓ Writes the plain-language analyst report.
- ✓ Performs structured MITRE ATT&CK Mobile mapping based exclusively on the deterministic engine's findings.

PS2: Evaluated the way a bank operates

9,082 accounts × 3,924 anonymised features, only 81 mules (0.9%).

Operational Metrics

~32× lift

Repeated stratified CV (5×5). Evaluated using PR-AUC and precision/recall constrained **strictly to a top-1% alert budget (the reality of a SOC team's daily capacity)**.

Feature Discovery

SHAP surfaces 30 predictive features entirely beyond the bank's 18 key inputs.

✓ Recommends promoting these new fraud-alert-like signals to **first-class** monitoring inputs.

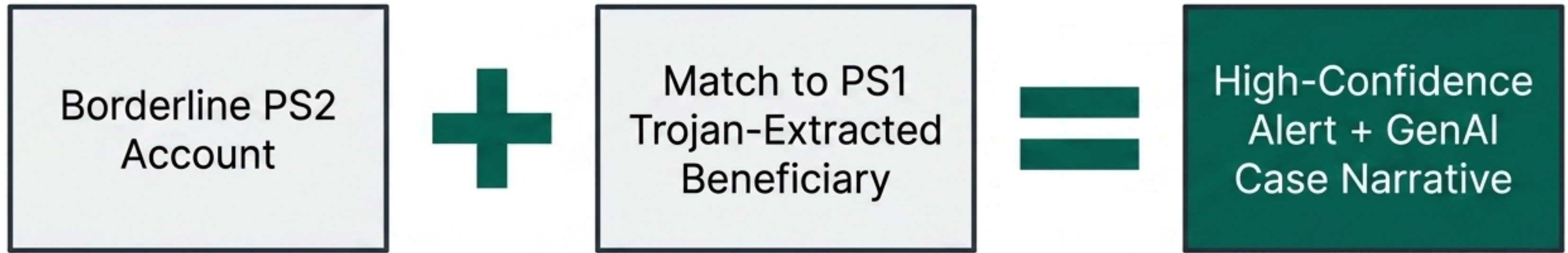
Dataset-Integrity Finding

A leakage audit revealed that feature F2230 (report month) perfectly confounds batch and label.

- All 9,001 negatives were extracted Oct-25; all 81 positives Sep/Nov/Dec-25.

Because of this confound, the reported performance metrics act as an honest upper bound.

The integration: the Threat-Intelligence Bus (upside, not load-bearing)



Each service stands alone and fully answers its own statement; the fusion is the upside. Demonstrated live today via the system's /case endpoint.

What runs today + roadmap

Working Today

- Live FastAPI service (/score-account with SHAP explanations, /case for fusion).
- Reproducible PS2 train/eval pipeline.
- 4 linted YARA-X rules.
- Public locally-runnable repo (github.com/PP112004/cybershield).

Prototype Phase (Jul-Aug 2026)

- Corpus-calibrated PS1 detection.
- NCRP/I4C free-text ticket parsing.
- Hosted public demo.

Limitations we state loudly: The report-month confound acts as an upper bound on metrics; initial PS1 risk weights are pre-calibration; the fusion is shown via anonymised join data.